

Skill-8

1. To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

```
ananya@DESKTOP-42I5F93:~$ cd skill8
ananya@DESKTOP-42I5F93:~/skill8$ pwd
/home/ananya/skill8
ananya@DESKTOP-42I5F93:~/skill8$ nano variable_expand.c
ananya@DESKTOP-42I5F93:~/skill8$ gcc variable_expand.c -o variable_expand
ananya@DESKTOP-42I5F93:~/skill8$ ./variable_expand
Skill-8 Variable Expansion
Enter text: My home is $HOME
Original : My home is $HOME
Expanded : My home is /home/ananya
ananya@DESKTOP-42I5F93:~/skill8$ ./variable_expand
Skill-8 Variable Expansion
Enter text: Home directory: ${HOME}
Original : Home directory: ${HOME}
Expanded : Home directory: /home/ananya
ananya@DESKTOP-42I5F93:~/skill8$ ./variable_expand
Skill-8 Variable Expansion
Enter text: Current user: $USER
Original : Current user: $USER
Expanded : Current user: ananya
ananya@DESKTOP-42I5F93:~/skill8$ ./variable_expand
Skill-8 Variable Expansion
Enter text: ./variable_expand
Original : ./variable_expand
Expanded : ./variable_expand
ananya@DESKTOP-42I5F93:~/skill8$ ./variable_expand
Skill-8 Variable Expansion
Enter text: Value: $XYZ123
Original : Value: $XYZ123
Expanded : Value: $XYZ123
ananya@DESKTOP-42I5F93:~/skill8$ ./variable_expand
Skill-8 Variable Expansion
Enter text: User=$USER Home=$HOME
Original : User=$USER Home=$HOME
Expanded : User=ananya Home=/home/ananya
ananya@DESKTOP-42I5F93:~/skill8$ ./variable_expand
Skill-8 Variable Expansion
Enter text: Welcome $USER to $HOME directory
Original : Welcome $USER to $HOME directory
Expanded : Welcome ananya to /home/ananya directory
```

```
ananya@DESKTOP-42I5F93:~/skill8$ nano builtin.c
ananya@DESKTOP-42I5F93:~/skill8$ gcc builtin.c -o builtin
ananya@DESKTOP-42I5F93:~/skill8$ ./builtin
Skill-8 Built-in Command Dispatcher
skill8$ pwd
/home/ananya/skill8
skill8$ echo Hello Skill 8
Hello Skill 8
skill8$ help

Built-in commands:
  cd <directory> - Change directory
  pwd            - Show current directory
  echo <text>    - Display text
  help          - Show help
  exit          - Exit shell
skill8$ ^C
ananya@DESKTOP-42I5F93:~/skill8$ ./builtin
Skill-8 Built-in Command Dispatcher
skill8$ pwd
/home/ananya/skill8
skill8$ help

Built-in commands:
  cd <directory> - Change directory
  pwd            - Show current directory
  echo <text>    - Display text
  help          - Show help
  exit          - Exit shell
skill8$ cd..
Invalid command: cd..
skill8$ pwd
/home/ananya/skill8
skill8$ cd ..
skill8$ pwd
/home/ananya
skill8$ cd skill8
skill8$ pwd
/home/ananya/skill8
skill8$ hello
Invalid command: hello
```

Built-in commands:

- cd <directory> - Change directory
- pwd - Show current directory
- echo <text> - Display text
- help - Show help
- exit - Exit shell

skill8\$ cd..

Invalid command: cd..

skill8\$ pwd

/home/ananya/skill8

skill8\$ cd ..

skill8\$ pwd

/home/ananya

skill8\$ cd skill8

skill8\$ pwd

/home/ananya/skill8

skill8\$ hello

Invalid command: hello

skill8\$ pwd

/home/ananya/skill8

skill8\$ cd ..

skill8\$ pwd

/home/ananya

skill8\$ exit

Exiting Skill-8 shell...